# Laxmi Charitable Trust's
# Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
## Department of Information Technology (B.Sc.I.T Semester V)
## Full Stack development MERN Practical
## Practical– VIII

| | |
|---|---|
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| Class: TYIT | Batch: 01 |
| Date of Assignment: : 30/09/2026 | Date/Time of Submission: 30/09/2026 |

### Practical 8: API Creation and Integration

**Aim:**

To create and integrate a REST API using Express, MongoDB and React.

### Explanation

- JavaScript normally works on a browser but with Node JS we can use it on server.
- Express is library runs on nodejs which makes API easy.
- Server.js creates express, allow react to communicate, read JSON data, connect to MongoDB, create API and start server.

### A. Create a Simple REST API using Express

**Steps:**

1. Create APIPractical folder and open it in VS Code.
2. Run **npm init -y.**

```
{
  "name": "apipractical",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "type": "commonjs"
}
```

3. Install mongoose, express and cors.

```
PS C:\Users\mvluc\Desktop\Siddhesh(T009)\MERN Practical\APIPractical> npm install mongoose express cors

added 88 packages, and audited 89 packages in 5s

30 packages are looking for funding
  run `npm fund` for details

found 0 vulnerabilities
```

4. Create **server.js.**

```
const express = require("express");
const mongoose = require("mongoose");
const cors = require("cors");

const app = express();
app.use(cors());
app.use(express.json());
mongoose.connect("mongodb://127.0.0.1:27017/college")
  .then(() => console.log("MongoDB Connected"));

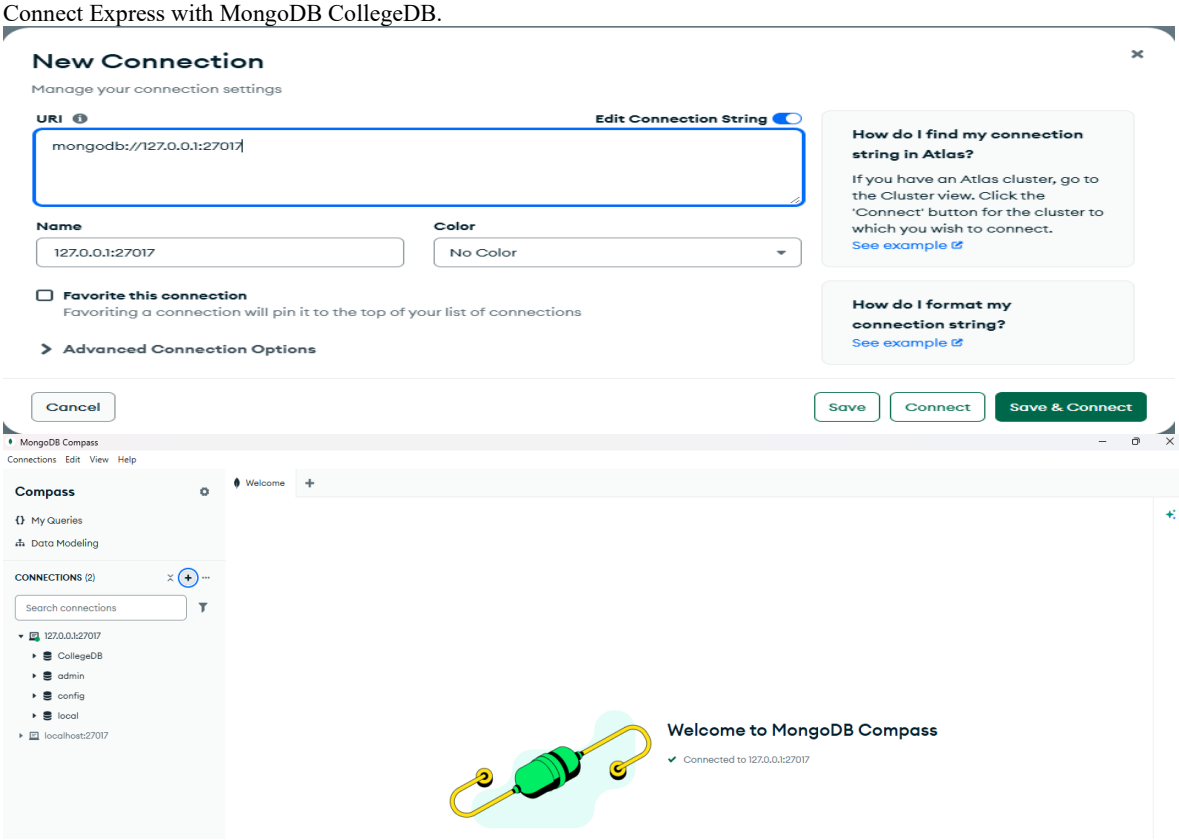
const Order = mongoose.model("students", new mongoose.Schema({
  studentName: String,
  rollNo: Number,
  course: String,
  foodItem: String,
  price: Number,
  quantity: Number,
  total: Number
}));
```

```
const food = [
  { name: "Pizza", price: 100 },
  { name: "Veg Burger", price: 60 },
  { name: "Sandwich", price: 50 },
  { name: "Samosa", price: 20 }
];
// GET /food
app.get("/food", (req, res) => {
  res.json(food);
});
// POST /orders
app.post("/orders", async (req, res) => {
  const { studentName, rollNo, course, foodItem, quantity } = req.body;

  const item = food.find(f => f.name === foodItem);
  const total = item.price * quantity;

  const order = new Order({
    studentName,
    rollNo,
    course,
    foodItem,
    price: item.price,
    quantity,
    total
  });
  await order.save();
  res.json(order);
});
// GET /orders
app.get("/orders", async (req, res) => {
  const orders = await Order.find();
  res.json(orders);
});
app.listen(5000, () => console.log("Server running on 5000"));
```

5. Connect Express with MongoDB CollegeDB.




6. Create a simple GET route /.
7. Run node server.js.

8. 
```
Server is running on http://localhost:3000
MongoDB Connected Successfully
```
9. Open http://localhost:3000.

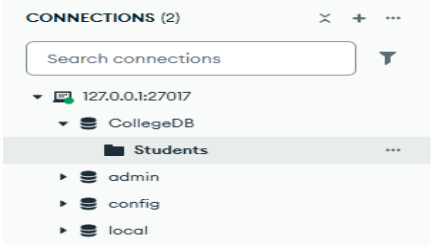
**Output:**




## B. Implement GET API

**Steps:**

1. Create CollegeDB database and students collection in MongoDB.




2. Insert student records.

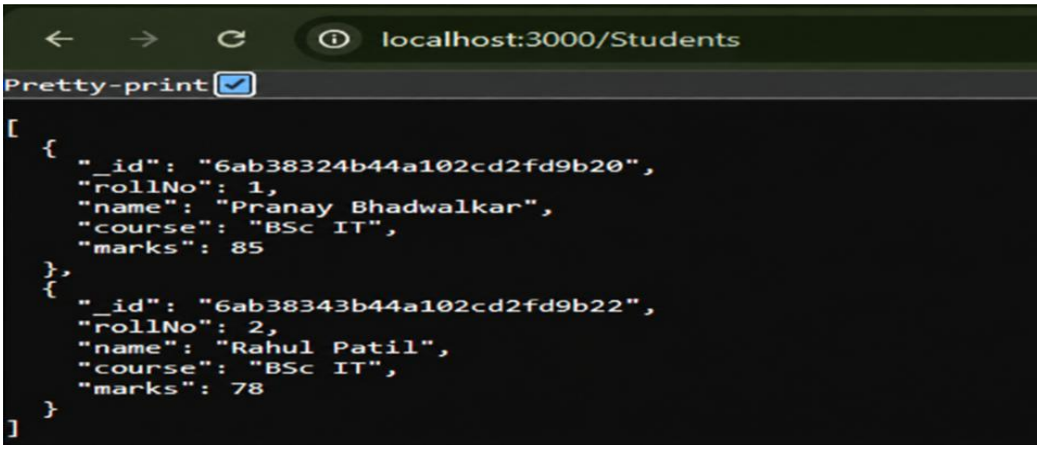



3. Add GET /students API in server.js.
   In `server.js`, add the following **above `app.listen()`**

```
app.get("/students", async (req, res) => {
    try {
        const students = await mongoose.connection.db
            .collection("students")
            .find()
            .toArray();
        res.json(students);
    }
    catch (error) {
        res.status(500).json({
            message: "Error fetching students"
        });
    }
});
```

4. Run the server.
5. Open http://localhost:3000/students.

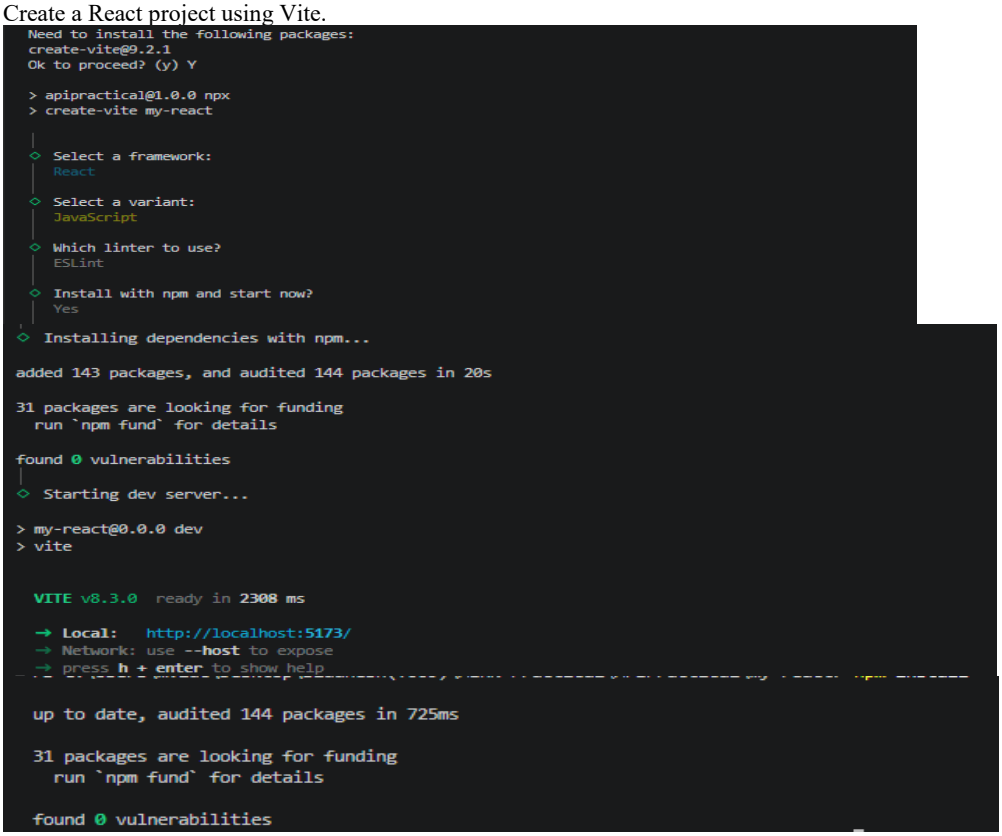
   **Output:**



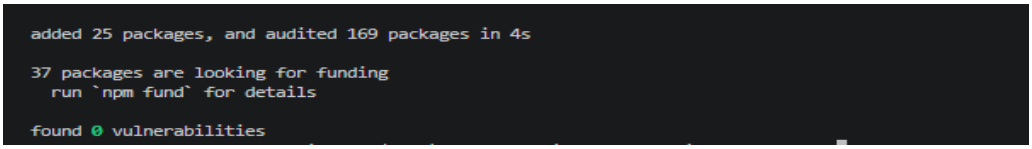

## C. Fetch API Data in React

**Steps:**

1. Create a React project using Vite.




2. Install Axios.



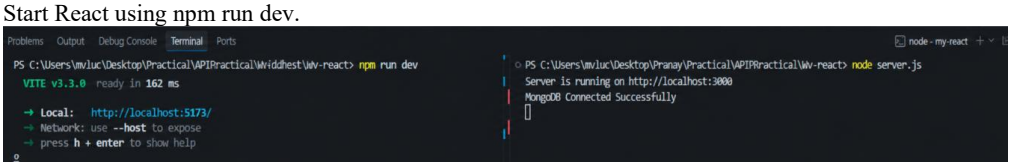

3. Add API request in App.jsx.

**App.jsx :**

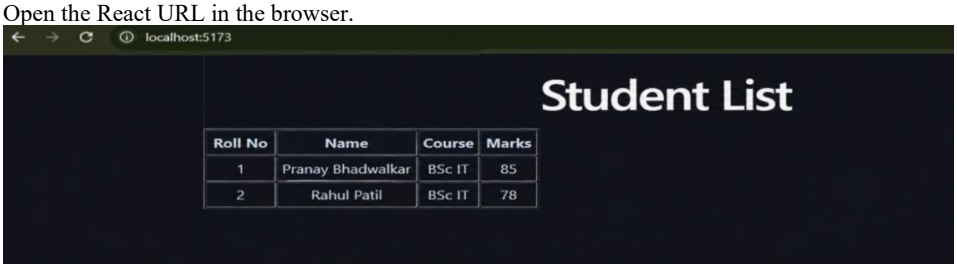
```
import { useEffect, useState } from "react";
import axios from "axios";
function App() {
    const [students, setStudents] = useState([]);
    useEffect(() => {
        axios.get("http://localhost:3000/students")
```

```
            .then((response) => {
                setStudents(response.data);
            })
            .catch((error) => {
                console.log(error);
            });
    }, []);
    return (
        <div>
            <h1>Student List</h1>
            <table border="1">
                <thead>
                    <tr>
                        <th>Roll No</th>
                        <th>Name</th>
                        <th>Course</th>
                        <th>Marks</th>
                    </tr>
                </thead>
                <tbody>
                    {students.map((student) => (
                        <tr key={student._id}>
                            <td>{student.rollNo}</td>
                            <td>{student.name}</td>
                            <td>{student.course}</td>
                            <td>{student.marks}</td>
                        </tr>
                    ))}
                </tbody>
            </table>
        </div>
    );
}
export default App;
```

4. Start Express server using node server.js.
5. Start React using npm run dev.




6. Open the React URL in the browser.




## D. Send Data using POST Request

**Steps:**

1. Add POST /students API in server.js.
   Add this code **above `app.listen()`**

```
app.post("/students", async (req, res) => {
    try {
        const student = req.body;
        await mongoose.connection.db
            .collection("students")
            .insertOne(student);
        res.json({
            message: "Student added successfully",
            student: student
        });
    }
```

```
    catch (error) {
        res.status(500).json({
            message: "Error adding student"
        });
    }
});
```

**App.jsx :**

```
import { useEffect, useState } from "react";
import axios from "axios";

function App() {
    const [students, setStudents] = useState([]);
    const [rollNo, setRollNo] = useState("");
    const [name, setName] = useState("");
    const [course, setCourse] = useState("");
    const [marks, setMarks] = useState("");
    // Get students
    const getStudents = () => {
        axios.get("http://localhost:3000/students")
            .then((response) => {
                setStudents(response.data);
            })
            .catch((error) => {
                console.log(error);
            });
    };
    useEffect(() => {
        getStudents();
    }, []);
    // Add student
    const addStudent = () => {
        const newStudent = {
            rollNo: Number(rollNo),
            name: name,
            course: course,
            marks: Number(marks)
        };
        axios.post("http://localhost:3000/students", newStudent)
            .then((response) => {
                alert(response.data.message);
                setRollNo("");
                setName("");
                setCourse("");
                setMarks("");
                getStudents();
            })
            .catch((error) => {
                console.log(error);
            });
    };
    return (
        <div>
            <h1>Student Management System</h1>
            <h2>Add Student</h2>
            <input
                type="number"
                placeholder="Roll No"
                value={rollNo}
                onChange={(e) => setRollNo(e.target.value)}
            />
            <br /><br />
            <input
                type="text"
                placeholder="Student Name"
                value={name}
                onChange={(e) => setName(e.target.value)}
            />
            <br /><br />
            <input
                type="text"
                placeholder="Course"
                value={course}
                onChange={(e) => setCourse(e.target.value)}
            />
            <br /><br />
```

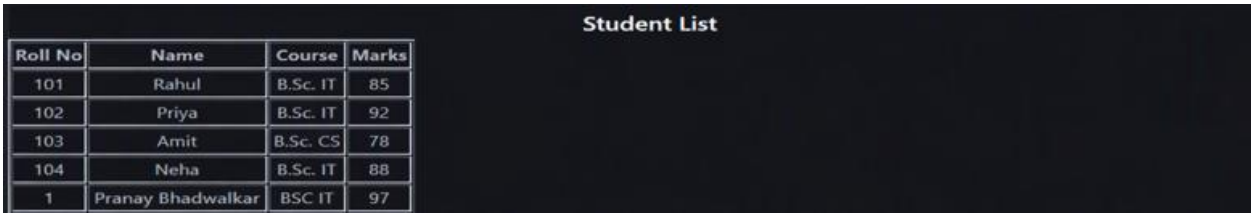
```
            <input
                type="number"
                placeholder="Marks"
                value={marks}
                onChange={(e) => setMarks(e.target.value)}
            />
            <br /><br />
            <button onClick={addStudent}>
                Add Student
            </button>
            <hr />
            <h2>Student List</h2>
            <table border="1">
                <thead>
                    <tr>
                        <th>Roll No</th>
                        <th>Name</th>
                        <th>Course</th>
                        <th>Marks</th>
                    </tr>
                </thead>
                <tbody>
                    {students.map((student) => (
                        <tr key={student._id}>
                            <td>{student.rollNo}</td>
                            <td>{student.name}</td>
                            <td>{student.course}</td>
                            <td>{student.marks}</td>
                        </tr>
                    ))}
                </tbody>
            </table>
        </div>
    );
}
export default App;
```

2. Create student input form in React.
3. Enter Roll No, Name, Course and Marks.
4. Click **Add Student**.
5. Axios sends the data to Express.
6. Express stores the data in MongoDB.
7. Student list is refreshed.

**Output:**



Name : Pranay Bhadwalkar Roll No. : T003




## Result

The REST API was successfully created using Express and MongoDB, and integrated with a React application using GET and POST requests.